

Inferenzstatistik

Zufallsvariable	Stichprobe
Zufallsvariable: X	Realisierungen: $x_1, x_2, x_3, x_4, \dots, x_n$
Wahrscheinlichkeitsverteilung $p(k)$ Wahrscheinlichkeitsdichte $f(X)$	Histogramm
Erwartungswert $E(X) = \mu$	Mittelwert $m = \bar{x}$ $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
Varianz $\text{Var}(X) = \sigma^2$	Probevarianz $\text{Var}(x_i) = s^2$

Inferenzstatistik

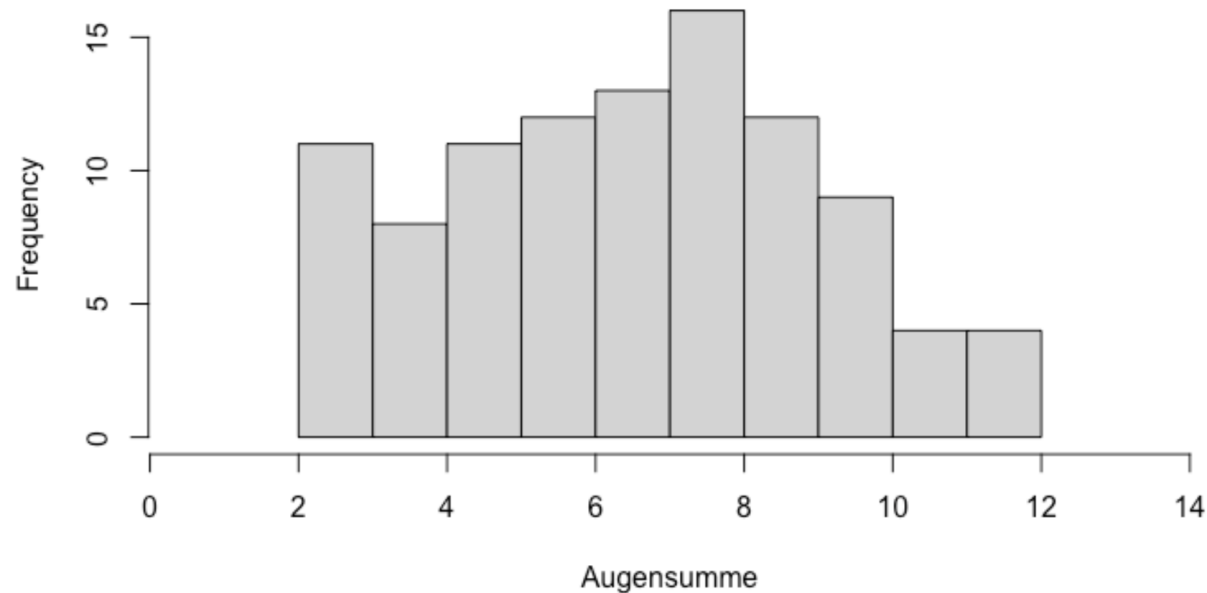
Können wir m und s benutzen
(Stichprobe), um μ und σ zu
schätzen (Zufallsvariable)?

Schätzung von μ

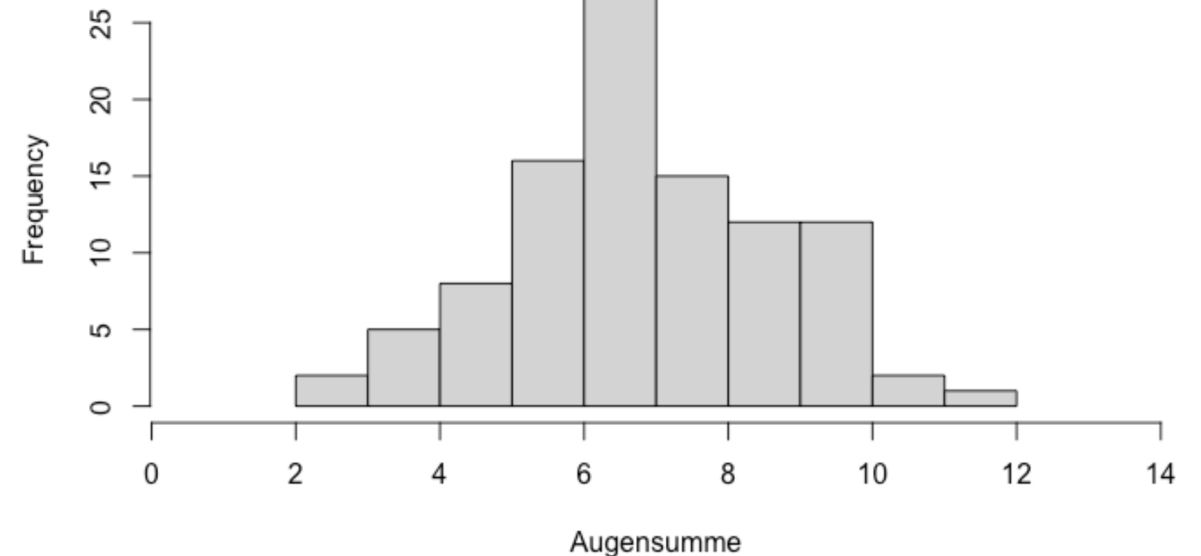
Experiment : z.B. ich berechne die Augensumme von 2 Würfeln.



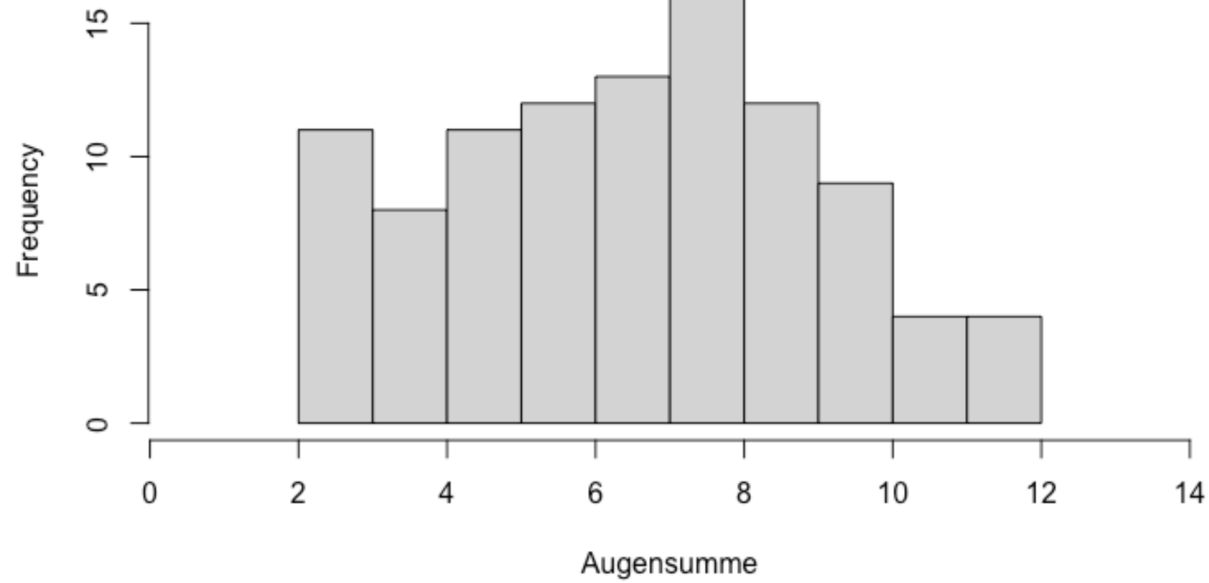
Augensumme for N = 1



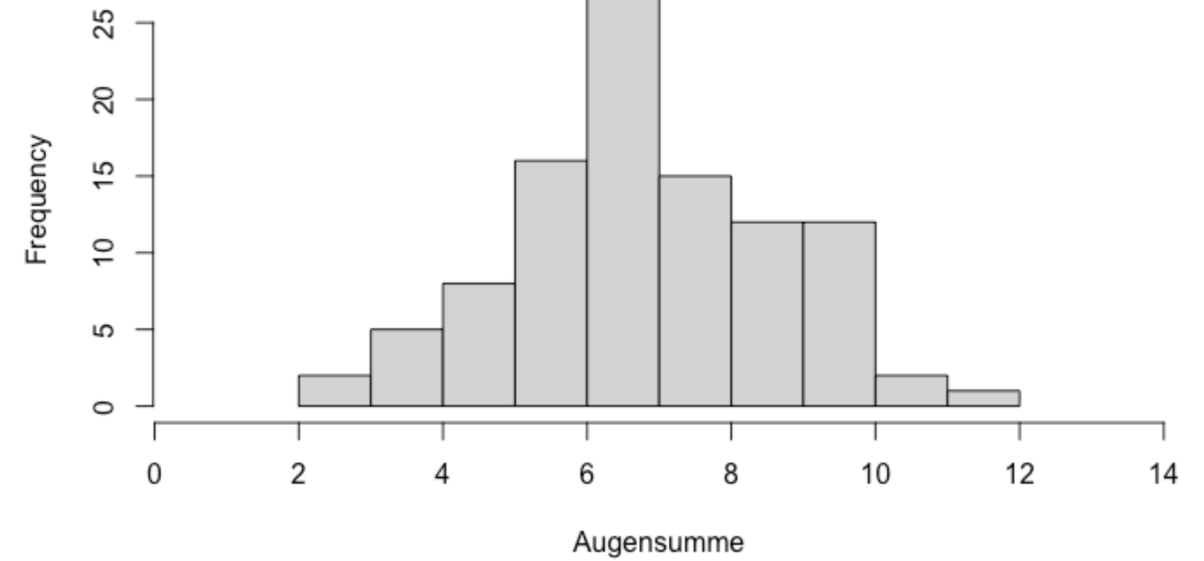
Augensumme for N = 2



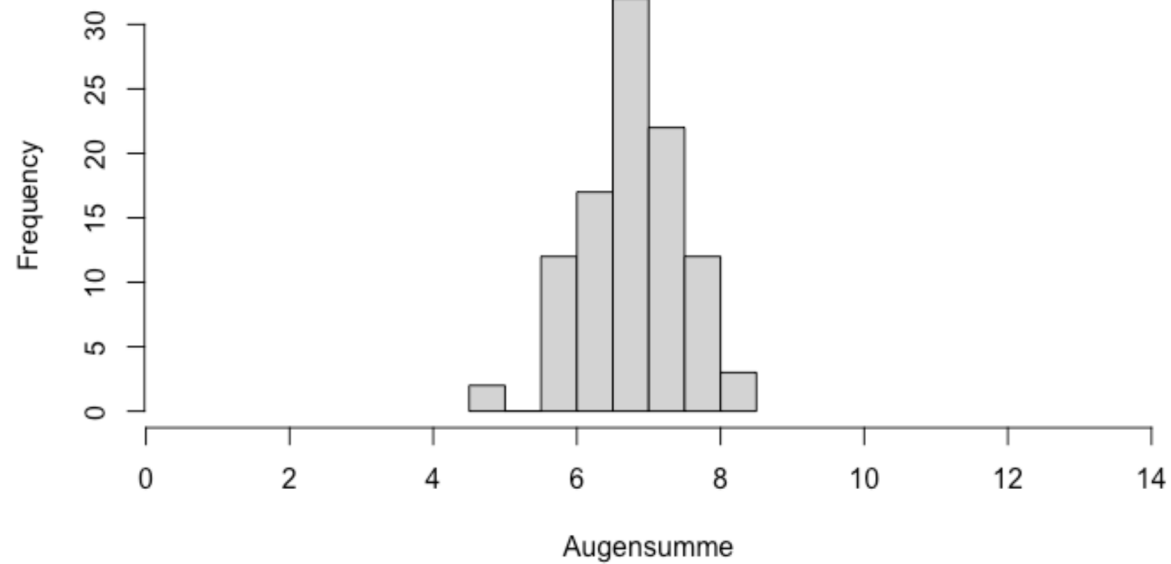
Augensumme for N = 1



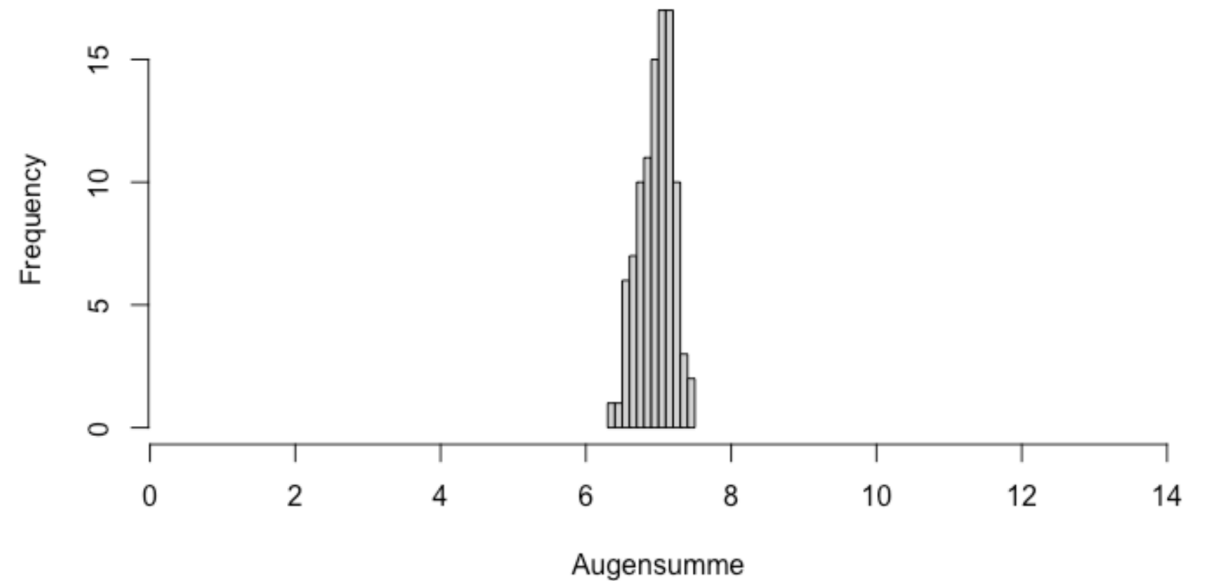
Augensumme for N = 2



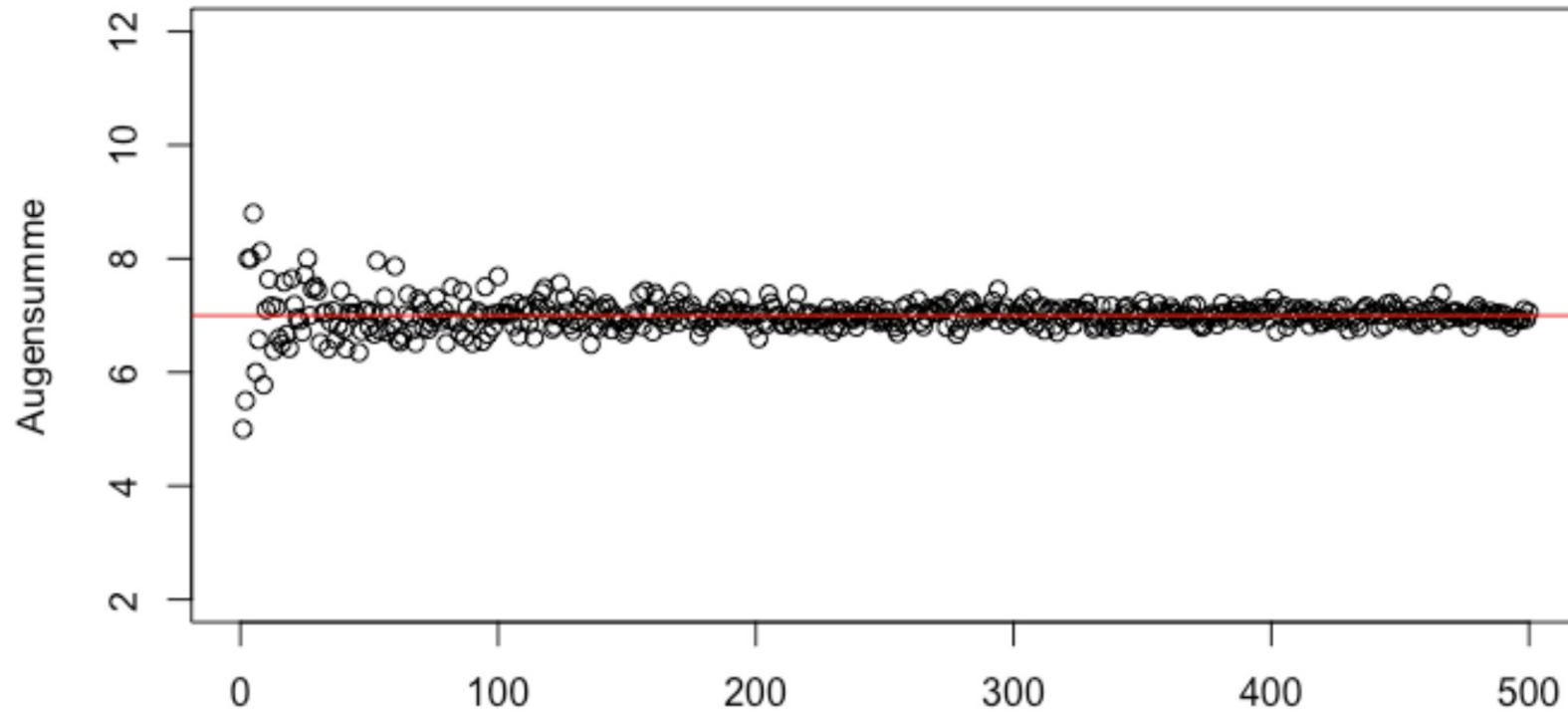
Augensumme for N = 10



Augensumme for N = 100



Mittelwert Augensumme wenn N größer wird



7 = Achtung Räuber !!!

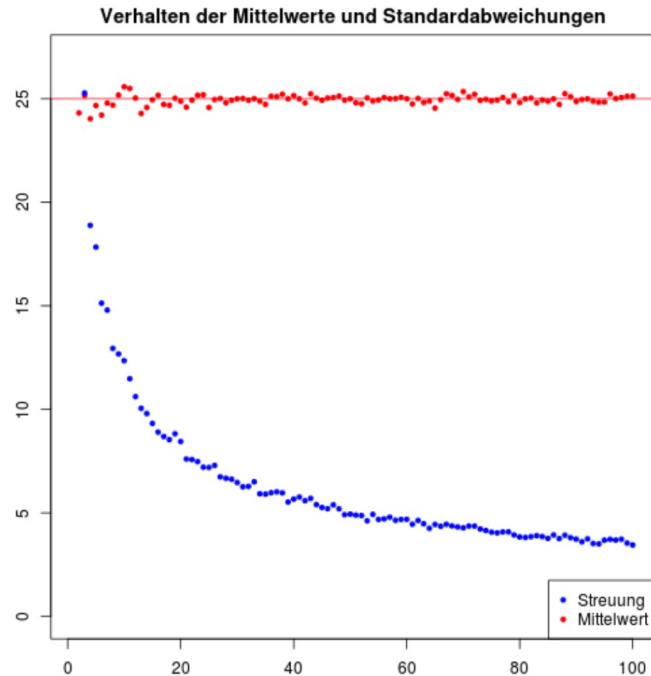


Der Mittelwert $\bar{m}$ der Stichprobe ist ein treuer Schätzer für den Erwartungswert μ der Zufallsvariable = der Mittelwert $\bar{m}$ der Stichprobe ist **erwartungstreu**

(unverzerrt = unbiased)

Estimating variance

$$Var_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \longrightarrow Var_n = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$



$$Var_n = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

is an **unbiased estimator** of the population variance

Was passiert wenn $N \gg 1$?

Zentraler Grenzwertsatz

- $\{X_1, X_2, \dots, X_n\}$ n random variables such that
 - independent
 - identically distributed  **any distribution !!!!**
 - $E[X_i] = \mu$; $Var(X_i) = \sigma^2$

- then the following holds:

Varianz

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{n \rightarrow \infty} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Standardabweichung

$$M_n \xrightarrow{n \gg 1} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- or:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{n \rightarrow \infty} \mathcal{N}(0,1)$$

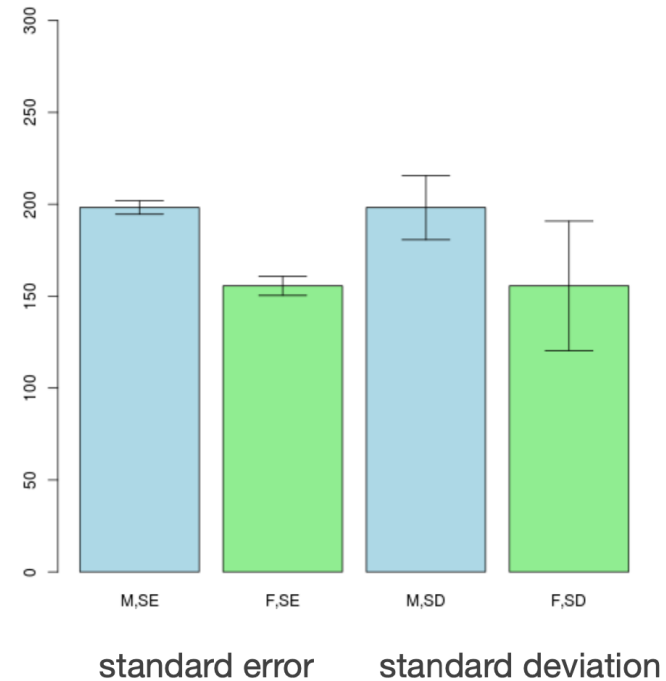
Standard error vs. Standard deviation

- **Standard deviation** describes the *spread in a distribution*

Standard deviation : σ

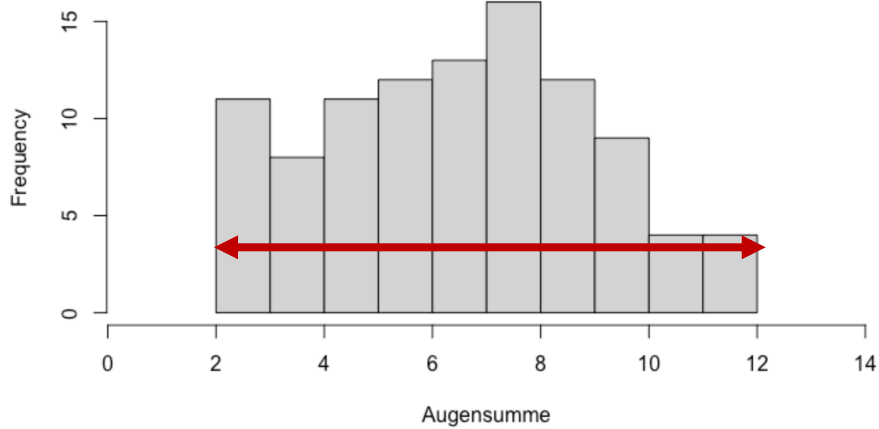
- **Standard error** measures the *spread in the estimation of the mean*

Standard error : $\frac{\sigma}{\sqrt{n}}$

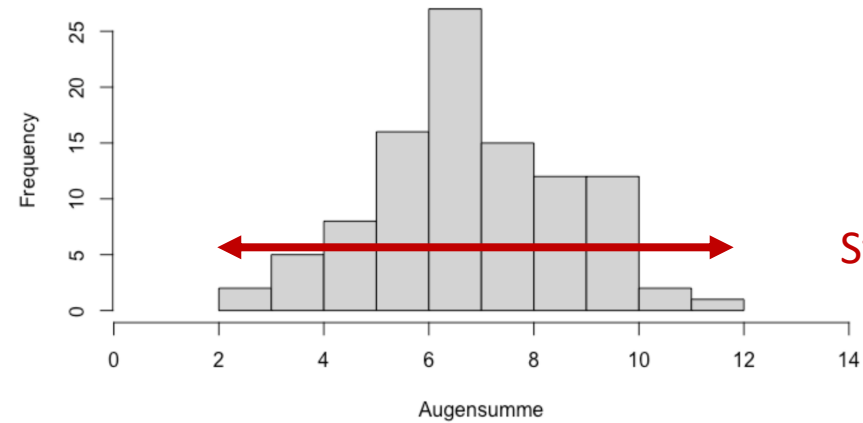


Zentraler Grenzwertsatz

Augensumme for N = 1

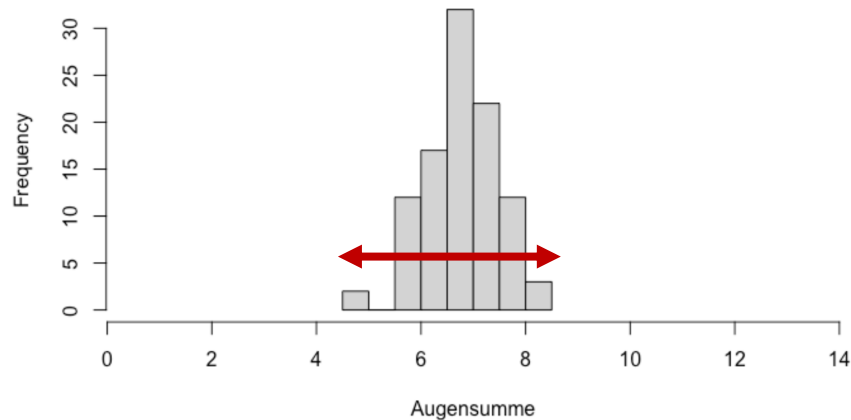


Augensumme for N = 2

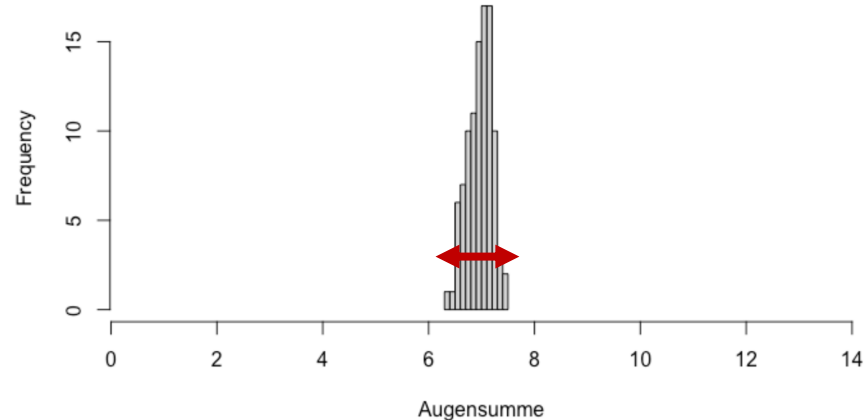


Streuung von den Mittelwerten

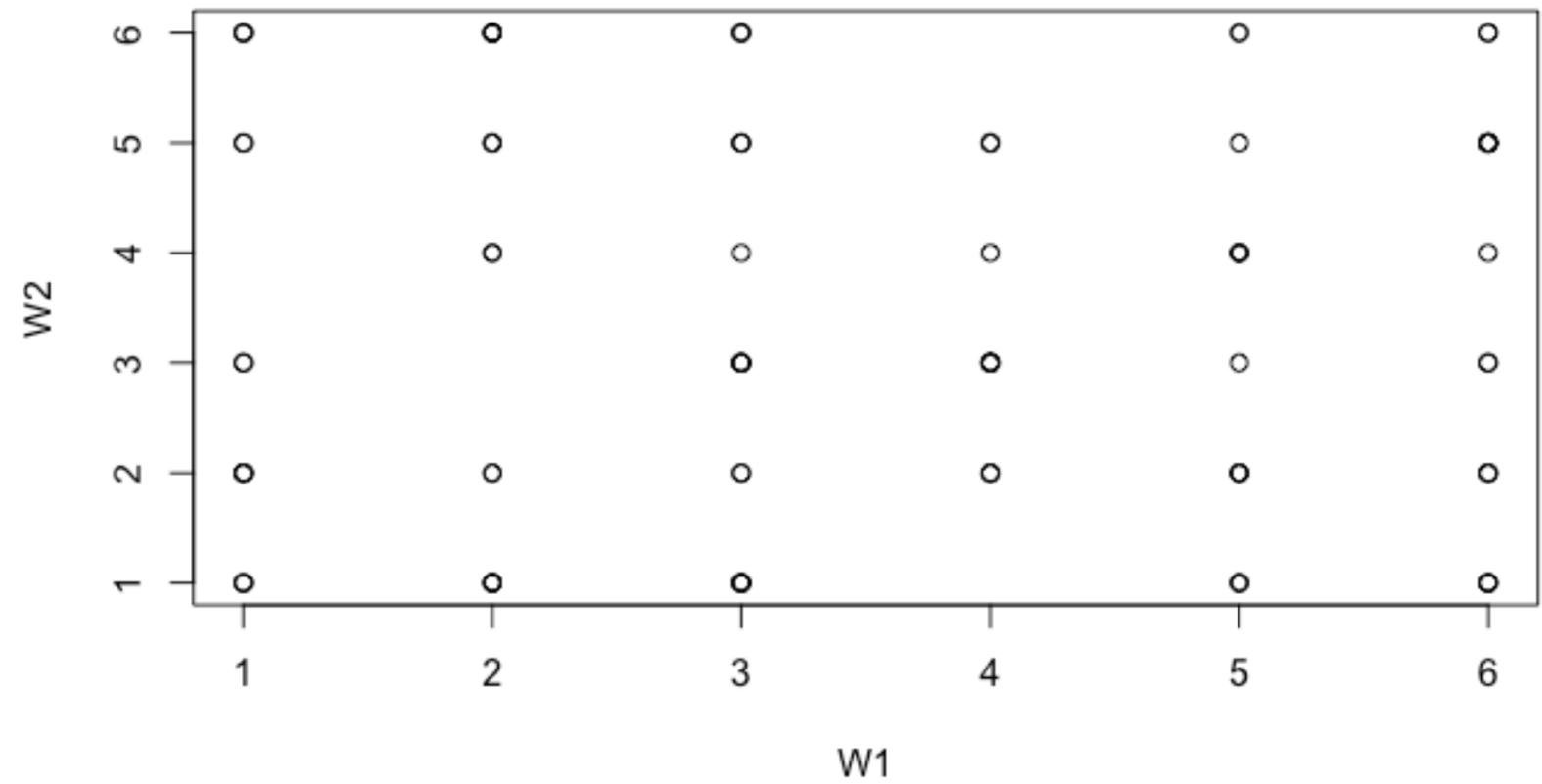
Augensumme for N = 10



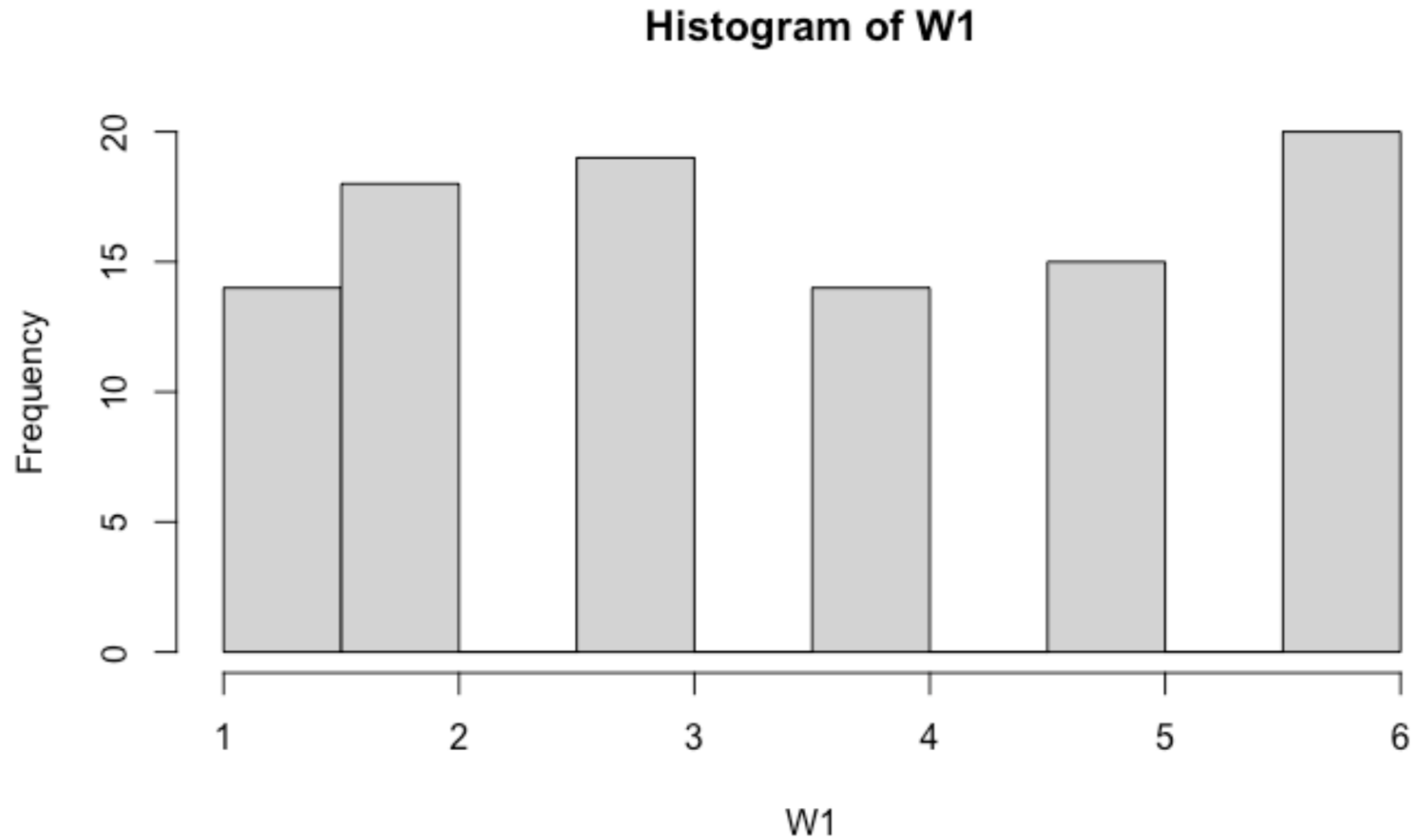
Augensumme for N = 100



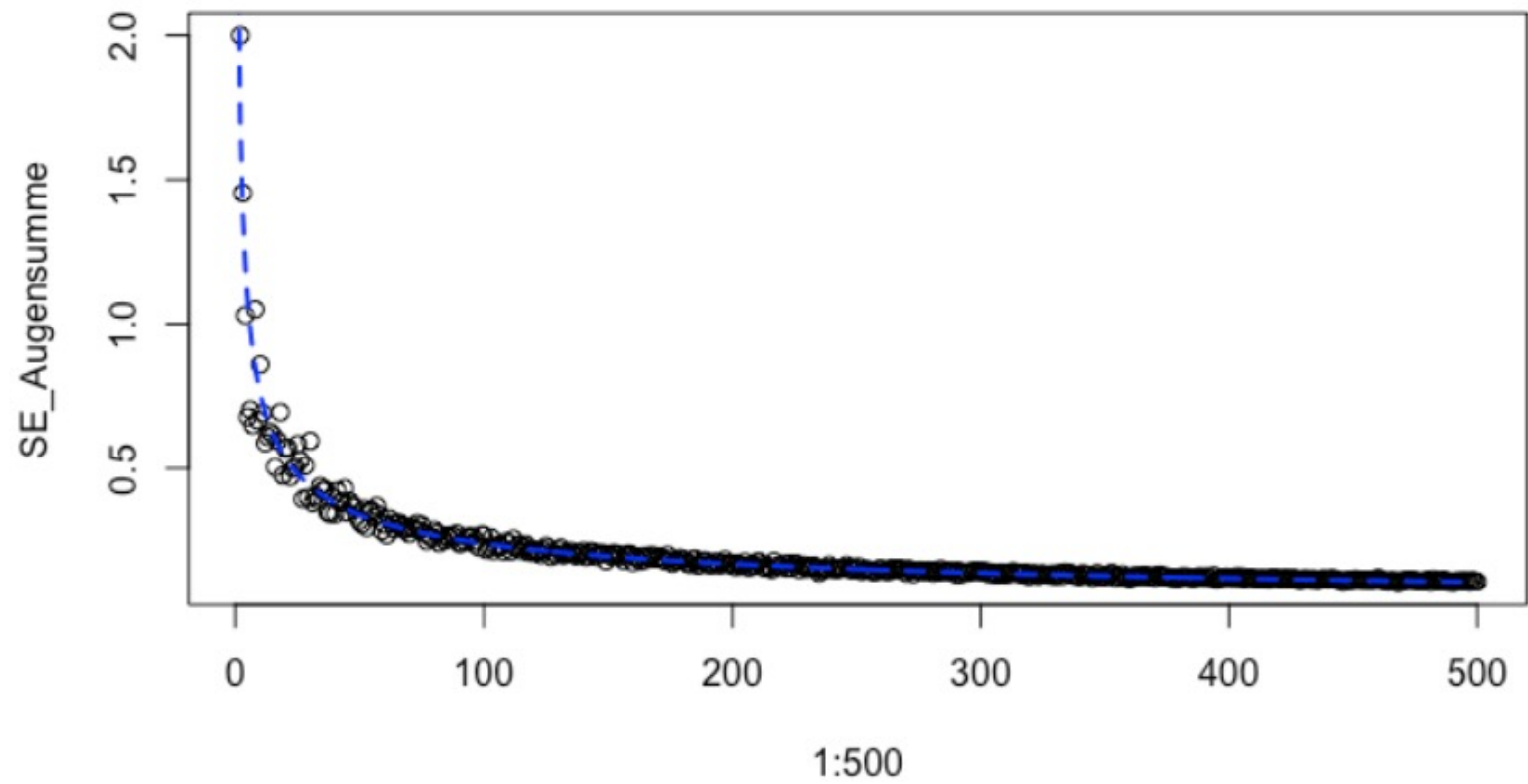
Unabhängig?



Aus der selben Verteilung?



SE wenn N größer wird



Konfidenzintervall

Confidence interval

$$M_n \xrightarrow{n \gg 1} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Central Limit Theorem

- large sample:

$$M_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right); \frac{M_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

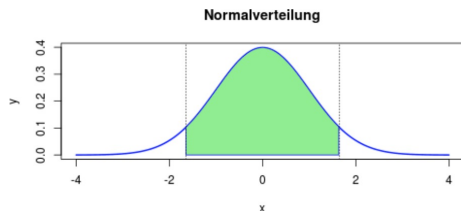
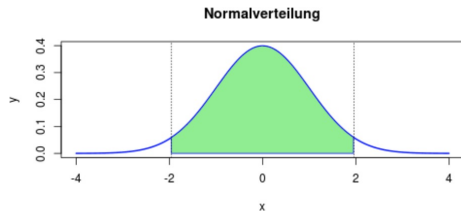
realization of the random variable!

- in 95% of the cases, $\left| \frac{m_n - \mu}{\sigma/\sqrt{n}} \right|$ is smaller than $t_{95}=1.96$
- 95% confidence interval:

$$\mu \in \left[m_n - t_{95} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95} \frac{\sigma}{\sqrt{n}} \right]$$

- Problem: we do not know the value of the population standard deviation σ
- Replace by its estimator

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$



Große Stichprobe -> SNV

Confidence interval

$$M_n \xrightarrow{n \gg 1} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- small sample:

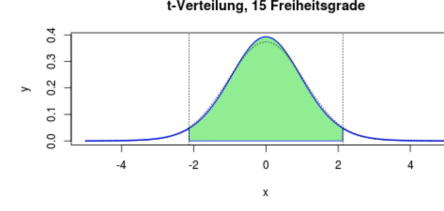
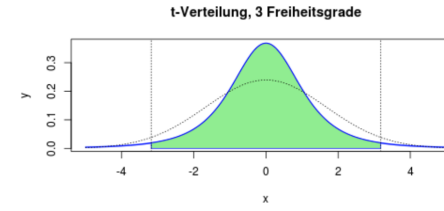
$$\frac{M_n - \mu}{\sigma/\sqrt{n}} \sim t_{n-1}$$

- in 95% of the cases, $\left| \frac{m_n - \mu}{\sigma/\sqrt{n}} \right|$ is smaller than $t_{95,n-1}$
- 95% confidence interval:

$$\mu \in \left[m_n - t_{95,n-1} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95,n-1} \frac{\sigma}{\sqrt{n}} \right]$$

- Problem: we do not know the value of the population standard deviation σ
- Replace by its estimator

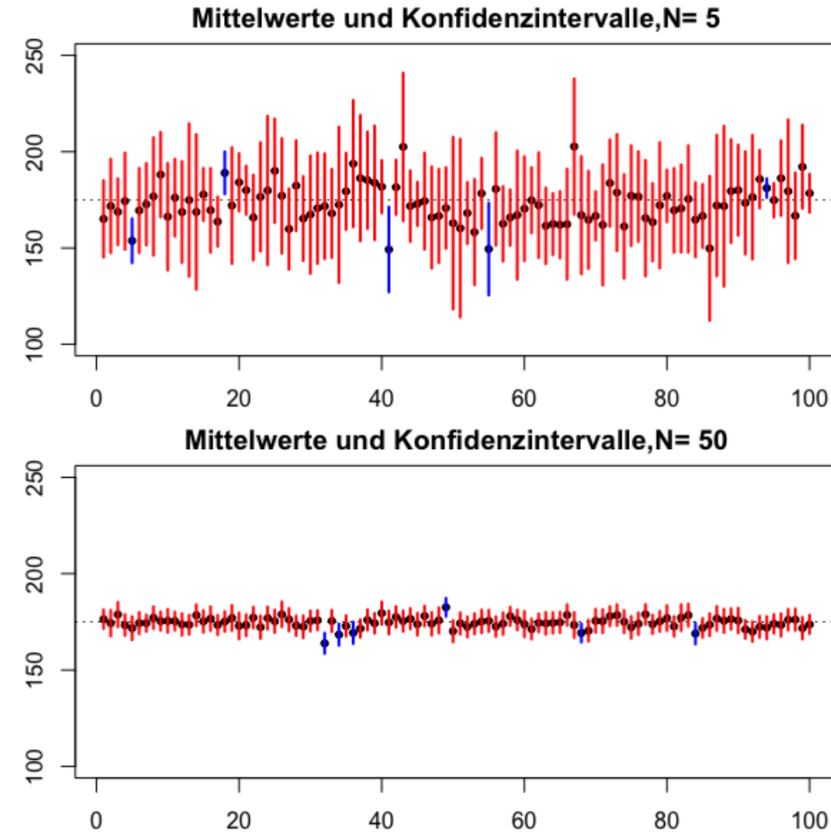
$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$



Kleine Stichprobe -> t-Verteilung

Interpretation of confidence interval

- 95% confidence interval
 - 100 samples with n elements
 - sample distribution with expectation value μ
 - estimate mean and CI
- True expectation value lies in 95% of the cases within the confidence interval



Confidence interval for Poisson distributions



- I find $k=20$ raisins in my raisin bun
→ what's in the recipe?
- Number of raisins N per bun is a **random variable**, described by a Poisson distribution

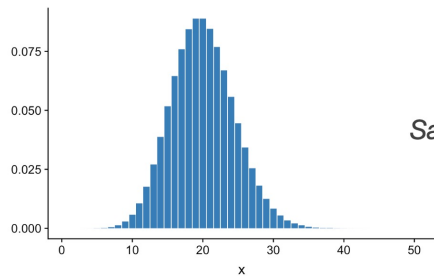
$$N \sim \text{Pois}(\lambda = k)$$

$$\text{Pois}(k) \xrightarrow{k \gg 1} \mathcal{N}(k, \sqrt{k})$$

Sample size $n=1$: $\mu \in [m_n - t_{95} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95} \frac{\sigma}{\sqrt{n}}]$

$$\mu \in [k - t_{95} \sqrt{k} ; k + t_{95} \sqrt{k}]$$

$$11.2 \leq \mu \leq 28.8$$



Confidence interval Poisson distribution

How to reduce the confidence interval?



28 17 22 19 18 ...

- **eat more buns!**
 $n=10$ samples with 1 bun
- Compute the mean over the sample $\bar{k} = 20.2$

$$\mu \in [m_n - t_{95} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95} \frac{\sigma}{\sqrt{n}}]$$

$$\mu \in [\bar{k} - t_{95} \frac{\sqrt{\bar{k}}}{\sqrt{10}} ; \bar{k} + t_{95} \frac{\sqrt{\bar{k}}}{\sqrt{10}}]$$

$$17.4 \leq \mu \leq 22.9$$

Week 5 - stop and think

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

- **Question 1:** According to the Central Limit Theorem, if we have n independent and identically distributed random variables $X_1, X_2, \dots, X_n$ with mean μ and variance σ^2 , the distribution of the sample mean $\bar{X}$ converges to:
 1. A uniform distribution with mean μ
 2. A Poisson distribution with parameter $\lambda = \mu$
 3. A normal distribution with mean μ and variance σ^2/n
 4. A t-distribution with $n-1$ degrees of freedom
- **Question 2:** Explain the difference between standard deviation and standard error. Why does the standard error decrease as sample size increases, and what does this tell us about our estimate of the population mean?
- **Question 3:** When calculating a 95% confidence interval for small samples, we use the t-distribution instead of the normal distribution. Looking at the critical values table in the lecture, explain what happens to the t-values as the sample size increases and why this makes sense.
- **Question 4:** Increasing the sample size will make the confidence interval smaller or larger? The 90% CI is larger or smaller than the 95% CI?

Critical values for t-distribution

df	t ₉₅	t ₉₀
2	4.302653	2.919986
3	3.182446	2.353363
4	2.776445	2.131847
5	2.570582	2.015048
8	2.306004	1.859548
10	2.228139	1.812461
15	2.131450	1.753050
20	2.085963	1.724718
50	2.008559	1.675905
100	1.983972	1.660234
200	1.971896	1.652508
Normal	1.959964	1.644854